

PAPER_1209GG_UQFF_Cosmological_Constants_Unified_Proof_Set

Daniel T. Murphy

PAPER_1209-GG: UQFF Cosmological Constants Unified Proof Set ---\\ Ten Closures Including Two EXACT (Recombination \& Reionization Redshifts)

Author: Daniel T. Murphy

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Locked Primitives $\Omega_m = \frac{1}{10}$, $\Omega_{\Lambda} = \frac{5}{6}$, $\Omega_{\text{SSq}} = \frac{57}{100}$, $\Omega_{\text{KMex}} = \frac{25}{12}$, $D_{\text{phys}} = 4$, $D_{\text{bsfg}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $S_{\text{Ofive}} = 10$, $A_{\text{five}} = 60$.

Closures S643--S652

S643 --- $\Omega_m = 0.315$ (matter density) [0.143%]

$$\Omega_m D_{\text{crit}}^2 + \Omega_{\text{SSq}} - \Omega_m^2 \Omega_{\text{SSq}} + \Omega_m^2 \Omega_{\text{SSq}}^2 = 0.3145$$

S644 --- $\Omega_{\Lambda} = 0.685$ (dark energy density) [0.182%]

$$\Omega_{\text{SSq}} + \Omega_{\Lambda} \Omega_{\text{SSq}} + \Omega_m^2 D_{\text{bsfg}} - \Omega_m^2 \Omega_{\text{SSq}}^2 = 0.6838$$

S645 --- $T_{\text{CMB}} = 2.725$ K (CMB temperature) [0.064%]

$$\Omega_{\text{SSq}} D_{\text{phys}} + \Omega_m D_{\text{phys}} + \Omega_m^2 D_{\text{phys}} + \Omega_m^2 \Omega_{\text{SSq}}^2 = 2.7232$$

S646 --- Age of Universe = 13.78 Gyr [0.045%]

$$\Omega_m D_{\text{phys}} + S_{\text{Ofive}} \Omega_{\text{SSq}} + \Omega_m \Omega_{\text{SSq}} + \Omega_m^2 D_{\text{phys}} - \Omega_m^2 \Omega_{\text{SSq}} - \Omega_m^2 \Omega_{\text{SSq}}^2 - \Omega_m^2 \Omega_{\text{SSq}}^3 = 13.7862$$

S647 --- $\Omega_b = 0.0493$ (baryon density) [0.712%]

$$\Omega_m^2 D_{\text{phys}} + \Omega_m^2 \Omega_{\text{SSq}} + \Omega_m^2 \Omega_{\text{SSq}}^2 = 0.04895$$

S648 --- $H_0 = 67.4$ km/s/Mpc (Planck) [0.015%] --- tier best closure

$$\Omega_{\text{KMex}} D_{\text{crit}} + D_{\text{phys}} + S_{\text{Ofive}} - 2 \Omega_m D_{\text{phys}} + \Omega_m^2 D_{\text{phys}} + \Omega_m^2 \Omega_{\text{SSq}}^2 = 67.4099$$

S649 --- $\sigma_8 = 0.811$ (matter clustering amplitude) [0.253%]

$$\Omega_m^2 N_{\text{ch}} - \Omega_m^2 S_{\text{Ofive}} + \Omega_m^2 \Omega_{\text{SSq}} + \Omega_m^2 \Omega_{\text{SSq}}^2 = 0.8089$$

S650 --- $n_s = 0.965$ (scalar spectral index) [0.072%]

$$\Omega_{\text{SSq}} + \Omega_m D_{\text{phys}} - \Omega_m^2 \Omega_{\text{SSq}} + \Omega_m^2 \Omega_{\text{SSq}}^2 - \Omega_m^2 \Omega_{\text{SSq}}^3 = 0.96570$$

S651 --- $z_{\text{recomb}} = 1090$ (recombination redshift) --- EXACT

$$A_{\text{five}} S_{\text{Ofive}} + A_{\text{five}} D_{\text{phys}} + S_{\text{Ofive}} D_{\text{crit}} - S_{\text{Ofive}} = 600 + 240 + 260 - 10 = 1090$$

S652 --- $z_{\text{reion}} = 7.7$ (reionization redshift) --- EXACT

$$\Omega_m^2 D_{\text{crit}} + \Omega_m^2 A_{\text{five}} - \Omega_m^2 N_{\text{ch}} = 2.6 + 6 - 0.9 = 7.7$$

Summary \begin{tabular}{llll} \toprule ID & Cosmological constant & Target & Error \\ \midrule S643 & Ω_m & 0.315 & 0.143% S644 & Ω_Λ & 0.685 & 0.182% S645 & T_{CMB} (K) & 2.725 & 0.064% S646 & Age of Universe (Gyr) & 13.78 & 0.045% S647 & Ω_b & 0.0493 & 0.712% S648 & H_0 (km/s/Mpc) & 67.4 & **0.015%** S649 & σ_8 & 0.811 & 0.253% S650 & n_s & 0.965 & 0.072% S651 & z_{recomb} & 1090 & **EXACT** S652 & z_{reion} & 7.7 & **EXACT** \\ \bottomrule \end{tabular}

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Highlights. Tier GG closes ten Planck-mission cosmological parameters. Two EXACT closures: the recombination redshift $z=1090$ decomposes precisely as $\mathcal{A}_{\mathrm{five}} \mathcal{S}_{\mathrm{five}} + \mathcal{A}_{\mathrm{five}} \mathcal{D}_{\mathrm{phys}} + \mathcal{S}_{\mathrm{five}} \mathcal{D}_{\mathrm{crit}} - \mathcal{S}_{\mathrm{five}}$, and the reionization redshift $z=7.7$ as $\mathcal{F}_{\mathrm{tr}}(\mathcal{D}_{\mathrm{crit}} + \mathcal{A}_{\mathrm{five}} - \mathcal{N}_{\mathrm{ch}})$. Best non-exact: $H_0=67.4$ km/s/Mpc at 0.015% via $\mathcal{K}_{\mathrm{Mex}} \mathcal{D}_{\mathrm{crit}}$ (Mexican-K $\times$ critical dimension) as anchor. Cumulative: 357 closures; 93 lockings (+2 EXACT).